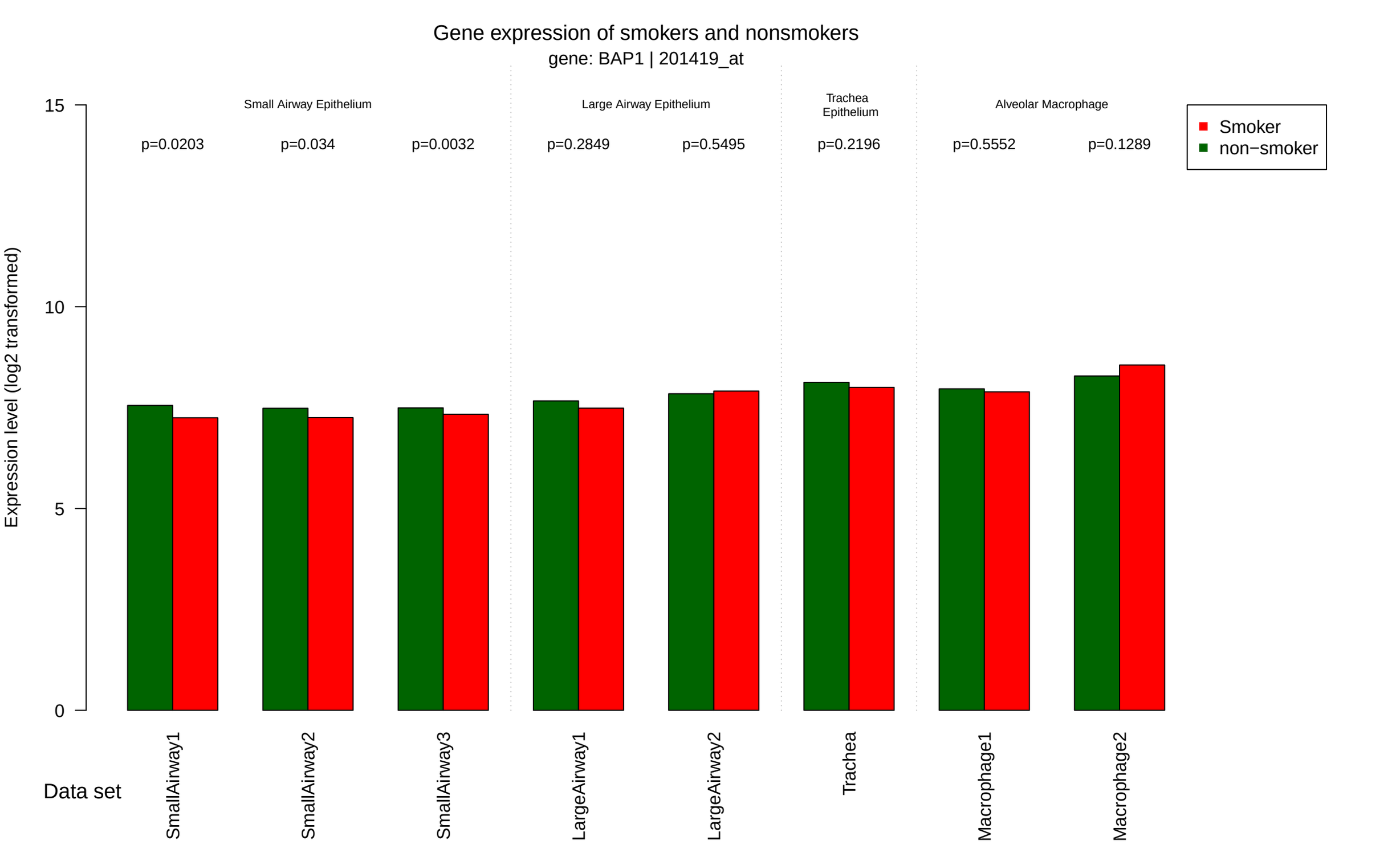
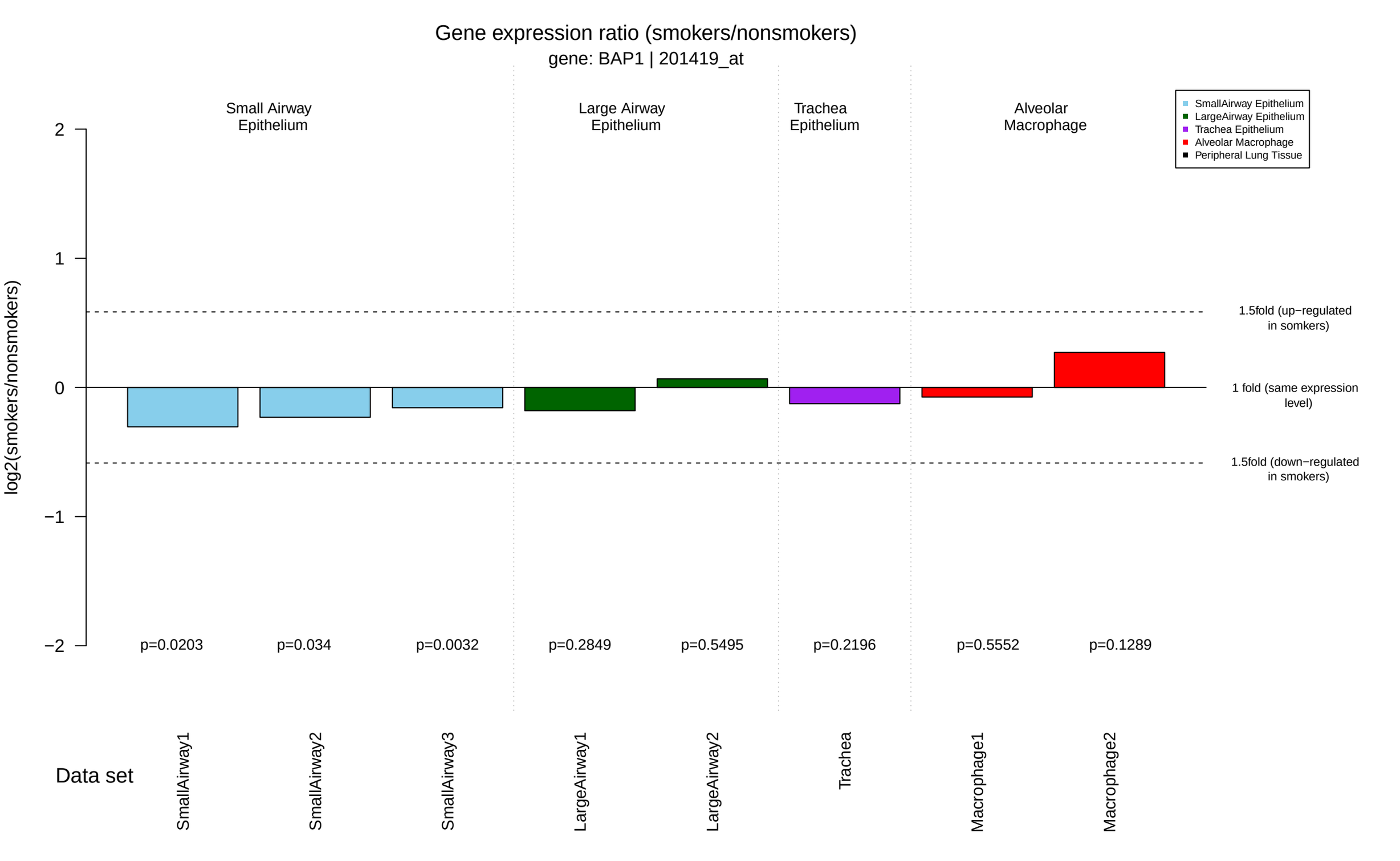
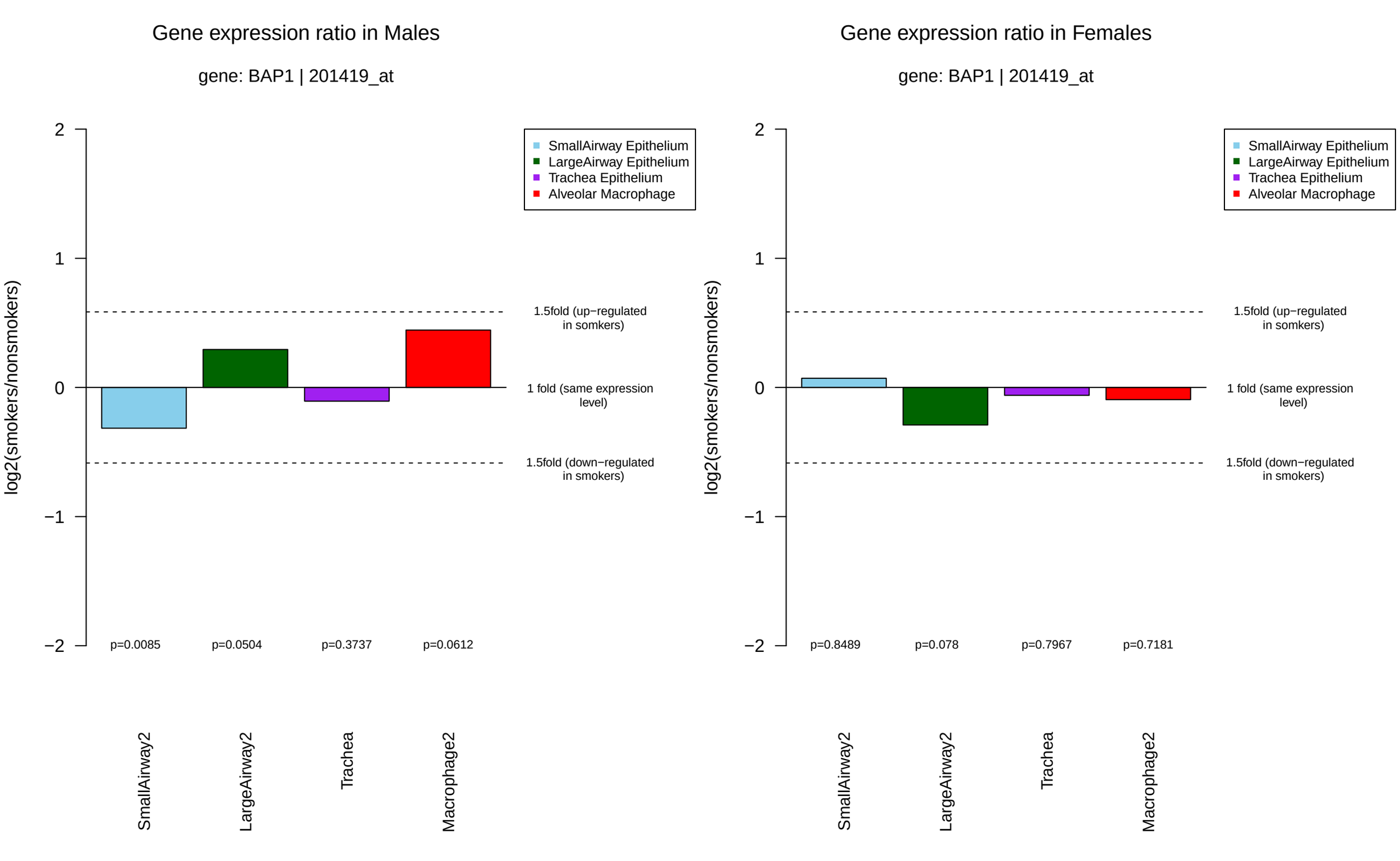
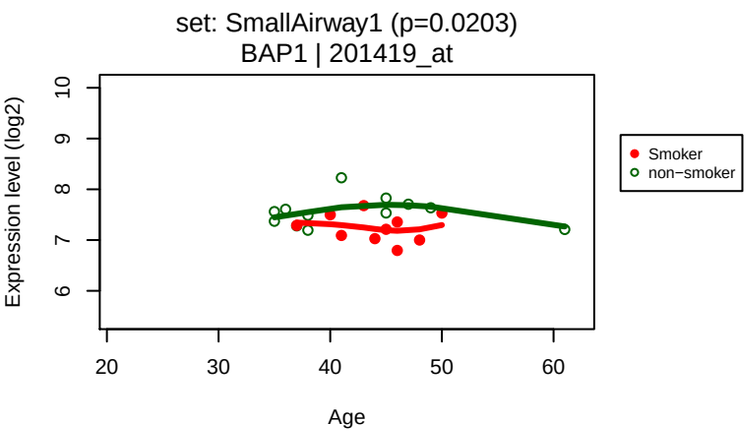
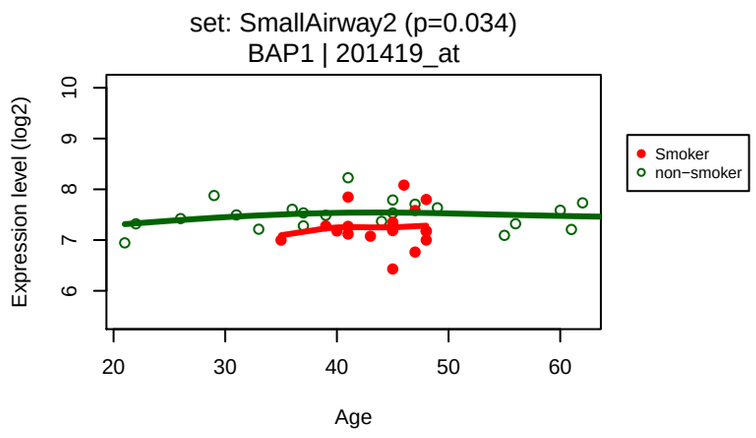
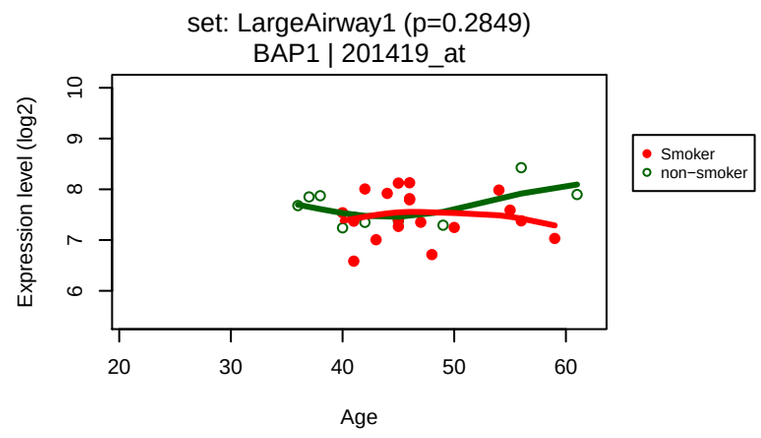

Gene expression of smokers and nonsmokers
gene: BAP1 | 201419_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0203
p=0.034
p=0.0032
p=0.2849
p=0.5495
p=0.2196
p=0.5552
p=0.1289
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

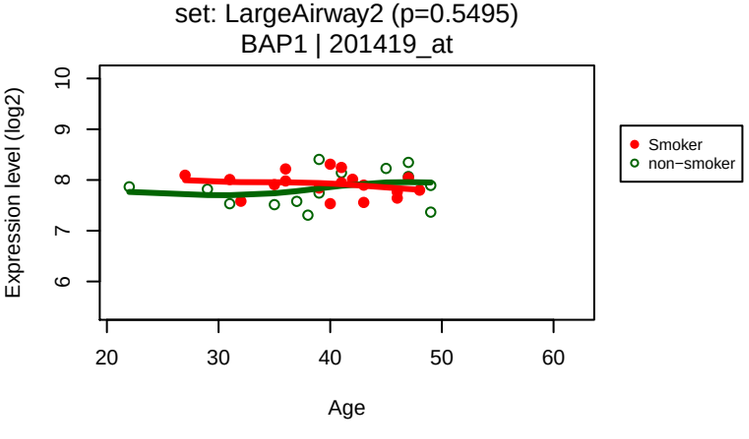

Gene expression ratio (smokers/nonsmokers)
gene: BAP1 | 201419_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0203
p=0.034
p=0.0032
p=0.2849
p=0.5495
p=0.2196
p=0.5552
p=0.1289
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

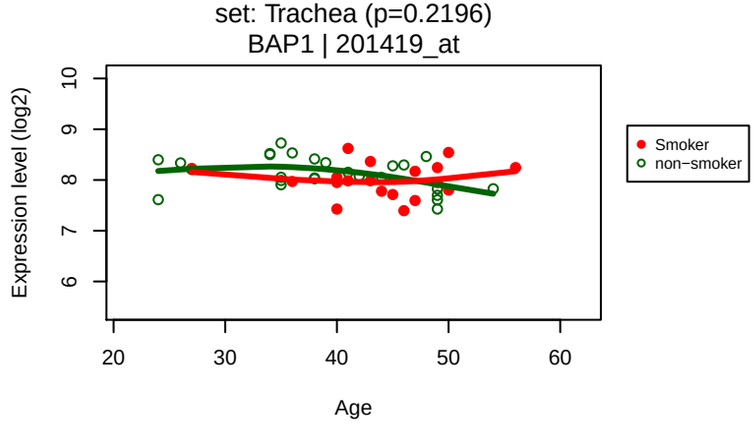

Gene expression ratio in Males
gene: BAP1 | 201419_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0085
p=0.0504
p=0.3737
p=0.0612
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: BAP1 | 201419_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8489
p=0.078
p=0.7967
p=0.7181
SmallAirway2
LargeAirway2
Trachea
Macrophage2

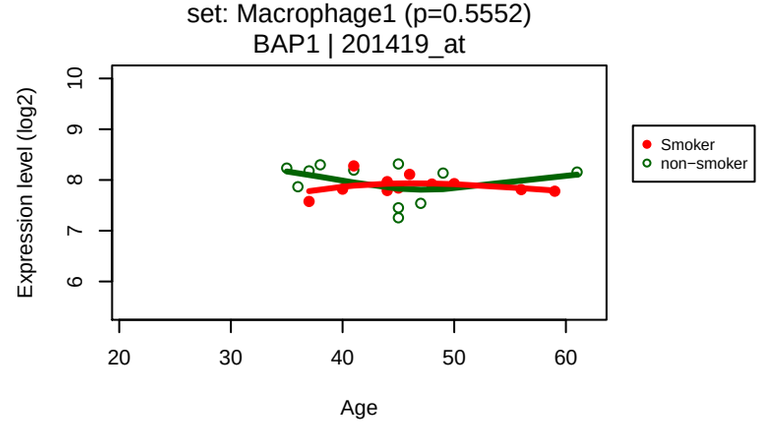

set: SmallAirway1 (p=0.0203)
BAP1 | 201419_at
Expression level (log2)
Age
Smoker
non−smoker

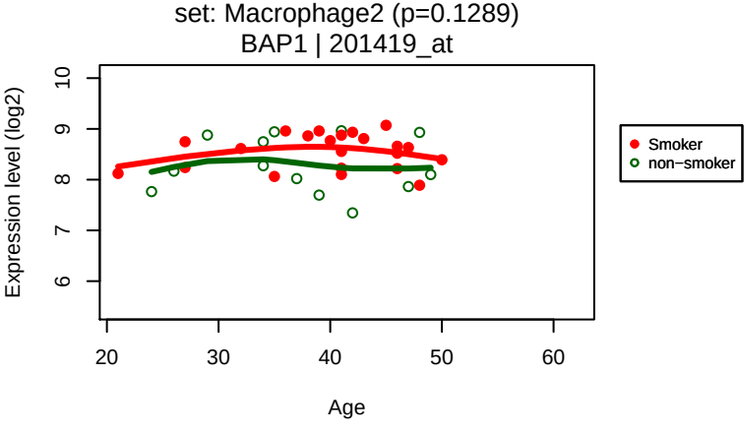

set: SmallAirway2 (p=0.034)
BAP1 | 201419_at
Expression level (log2)
Age
Smoker
non−smoker

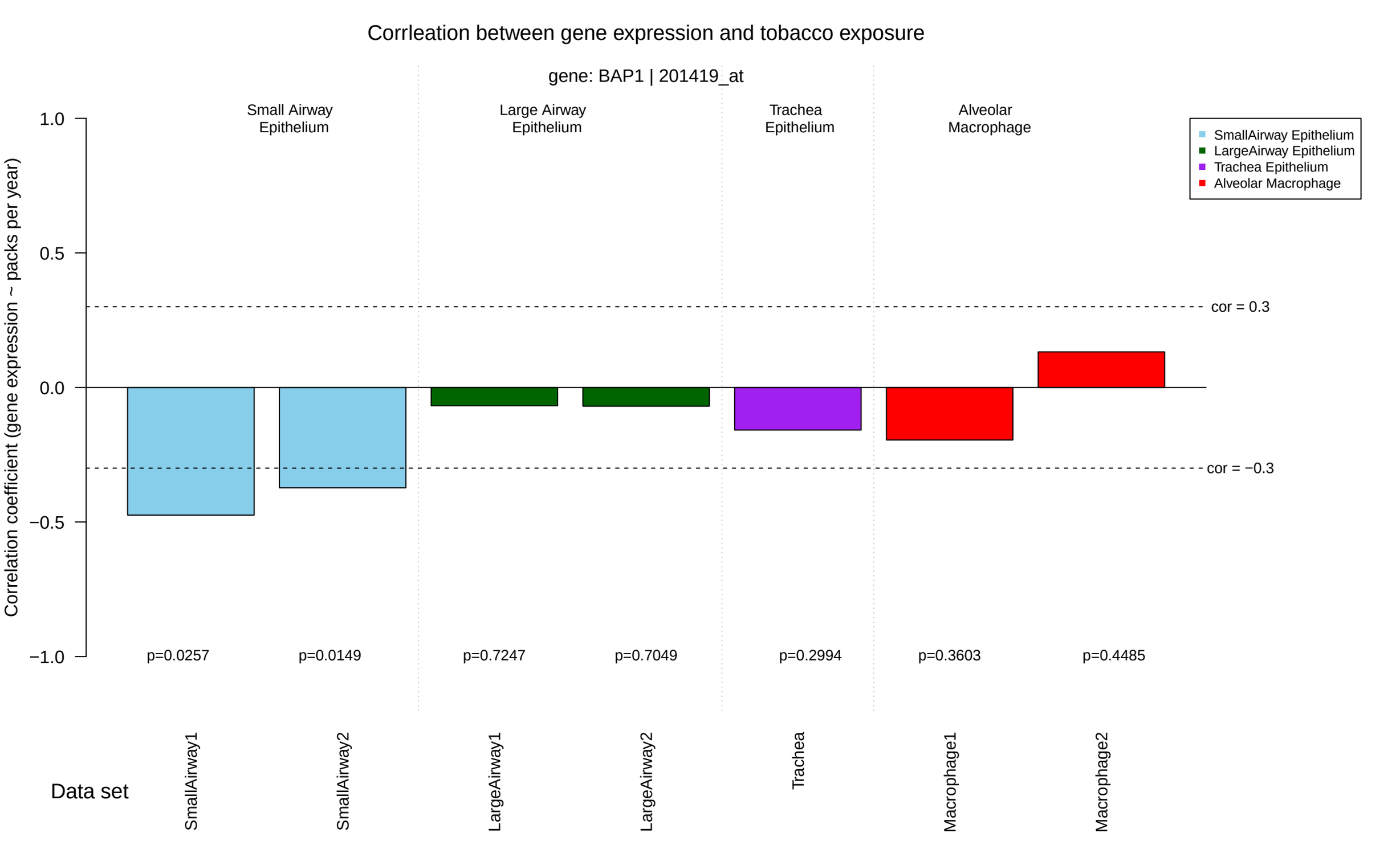

set: LargeAirway1 (p=0.2849)
BAP1 | 201419_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5495)
BAP1 | 201419_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2196)
BAP1 | 201419_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5552)
BAP1 | 201419_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1289)
BAP1 | 201419_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: BAP1 | 201419_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0257
p=0.0149
p=0.7247
p=0.7049
p=0.2994
p=0.3603
p=0.4485
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2